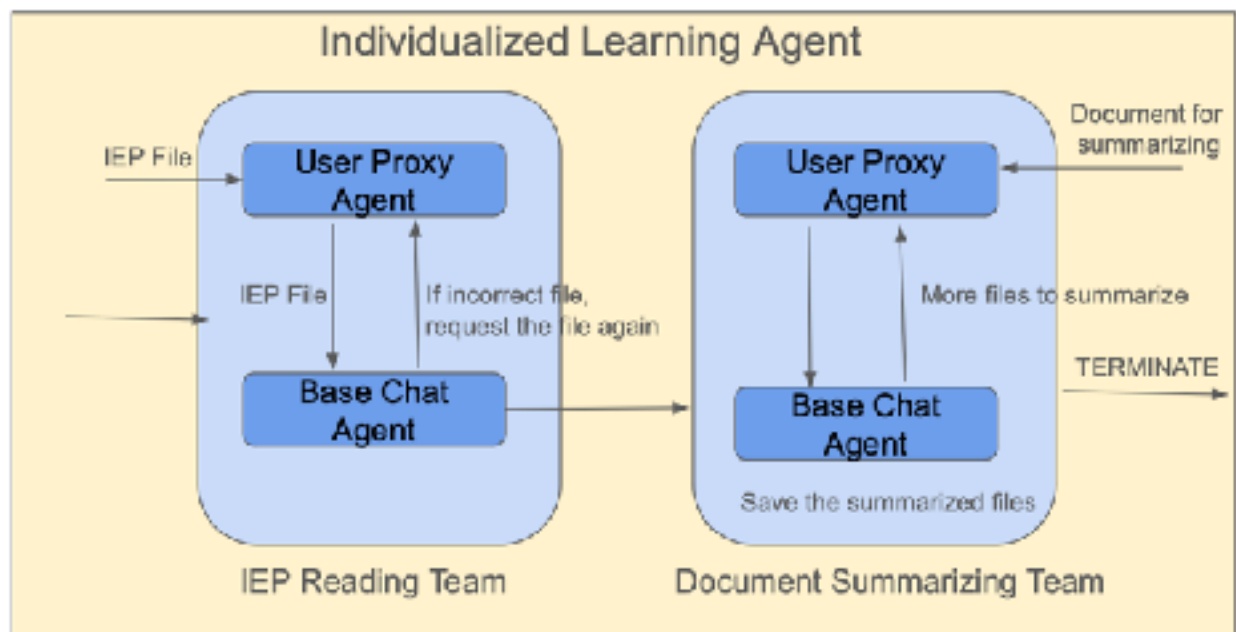


Project Description: Individualized Learning Agent

The **Individualized Learning Agent** is designed to ensure that no student is left behind, enabling every learner to thrive according to their unique capabilities. This application leverages the power of **Azure OpenAI** and a **multi-agent architecture** to adapt educational content to individual learning levels. By analyzing a student's comprehension ability, the system intelligently rewrites textbook chapters or classroom documents to align with the student's cognitive level. This personalized approach fosters deeper understanding, increased engagement, and improved knowledge retention.

Architecture Overview



This is an **AutoGen-based application** utilizing two collaborative agent teams:

- **Team 1** engages with the user and gathers the student's learning level via a BaseChatAgent.
- **Team 2** takes input from the user and processes PDF documents through another BaseChatAgent, generating simplified summaries tailored to the learner's ability.

The adapted content is automatically saved in the same folder as the original PDF for easy access and reference.

Target Users

Educators, Parents, and Students — particularly those working with students who have special needs or learning differences. This tool can be integrated into existing educational platforms or used independently to provide a truly **individualized learning experience**. It empowers educators to adapt materials to meet each learner where they are, enhancing inclusivity and learning outcomes in both traditional and special education settings.